The empirical recurrence for OEIS A063116, proved from an exact model

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2 September 2026

Abstract

OEIS A063116 carries an empirical recurrence of order 2 contributed by David James Sycamore. It is true, and it is decidable rather than empirical. The entry's sequence is given exactly by the classical dimension formula for cusp forms, from which a provably satisfies a MONIC linear recurrence of order $S = 8$, derived below and not assumed. The residual of the conjectured recurrence satisfies the same one, so whether it annihilates a is settled by a finite exact computation. No transfer matrix is involved and the count is not a walk.

2020 Mathematics Subject Classification. 11F11, 11F72, 05A15.

1 The conjecture

OEIS A063116 is "Dimension of the space of weight 2n cusp forms for Gamma_0(48).". It has offset 1 and begins

$$3, 18, 34, 50, 66, 82, 98, 114, \ \dots$$

The entry states, as a conjecture contributed by its author and never marked settled:

> Conjecture: For n>=2, a(n) is the difference between the greater of successive pairs of odd triangular numbers(A000217), namely A(B(n-1)+1) - A(B(n-2)+1), where A=A000217 and B=A016813. This reduces to a(n)=2*(8*n-7) (see Formula). (1,3),(15,21) are the first consecutive pairs of odd triangular numbers, and 21-3=18, which is a(2). - _David James Sycamore_, Sep 12 2019

As of the "Last modified" line on the live entry (Sep 12 2019 12:33:29, revision 24) this is still recorded as empirical, and nothing on the entry records it as proved.

2 The dimension is given in closed form

There is no array here and no transfer matrix. For even $k \geq 4$ the dimension of the space of cusp forms is given exactly by Diamond and Shurman, Theorem 3.5.1,

$$\dim S_k(\Gamma_0(N)) = (k-1)(g-1) + \left(\tfrac{k}{2} - 1\right)\varepsilon_\infty + \left\lfloor \tfrac{k}{4} \right\rfloor \varepsilon_2 + \left\lfloor \tfrac{k}{3} \right\rfloor \varepsilon_3,$$

with $\dim S_2 = g$ and $\dim S_0 = 0$. Every quantity on the right is elementary integer arithmetic in N: the index $\mu = N \prod_{p|N}(1 + 1/p)$; the elliptic counts $\varepsilon_2, \varepsilon_3$, which vanish when $4 \mid N$ and $9 \mid N$ respectively and are otherwise products of $1 + \left(\frac{-1}{p}\right)$ and $1 + \left(\frac{-3}{p}\right)$ over the primes dividing N; the cusp count $\varepsilon_\infty = \sum_{d|N} \varphi(\gcd(d, N/d))$; and the genus $g = 1 + \mu/12 - \varepsilon_2/4 - \varepsilon_3/3 - \varepsilon_\infty/2$.

3 The bound

With $k = 2n$ the right-hand side is linear in n apart from $\lfloor n/2 \rfloor$ and $\lfloor 2n/3 \rfloor$, so for $n \geq 2$ — that is, for the even weights $k \geq 4$ the formula covers — a is a quasi-polynomial of period 6 and degree 1 satisfying $a(n+6) = a(n) + 6A + 3\varepsilon_2 + 4\varepsilon_3$ exactly. That difference equation is annihilated by $(z^6 - 1)(z - 1)$, and $S = 8$ is at least its degree.

The two remaining indices are genuinely outside it: $\dim S_0 = 0$ and $\dim S_2 = g$ are separate cases of the theorem, not values of the same expression, and the annihilator does not reach them — on every one of these entries the residual of $(z^6 - 1)(z - 1)$ is nonzero at $n = 0$ and $n = 1$ and vanishes from $n = 2$ on. That costs nothing here: $a(n + 2)$ is annihilated everywhere, the residual below is computed term by term from the closed form rather than through A, and the run of zeros the theorem uses lies entirely above the transient.

4 The proof

Lemma 1. *Let $A(z) = z^S - \sum_{j<S} \alpha_j z^j$ be the monic annihilator derived above and $q(z) = z^2 - \sum_i c_i z^{2-i}$ the characteristic polynomial of the conjectured recurrence. The residual $r(n) = a(n) - \sum_i c_i a(n - i)$ is a linear combination of shifts of a, so A annihilates r as well. Since $A(0) \neq 0$ the recurrence it gives runs in both directions, and S consecutive zeros of r force r to vanish at every later index.*

Proof. A annihilates a, hence every shift of a, hence any linear combination of shifts; that is r. A monic recurrence with nonzero constant term determines each term from the S before it and each from the S after it, so a run of S zeros propagates. □

Theorem 2.

$$a(n) \;=\; 2\,a(n-1) - a(n-2)$$

for every $n > 3$.

Proof. The terms $a(0), \ldots$ were computed exactly in integer arithmetic from the model, extended by A where the model itself was not evaluated, and $r(n)$ formed for each index. Those integers vanish from $n = 3+1$ onwards, and the run exceeds $S+2$, which by Lemma 1 settles every larger index at once. The bound is exact: at $n = 3$ the recurrence fails on the entry's own published terms, so no larger range holds. □

5 Verification

Three checks, each independent of the algebra above.

First, the model reproduces all 50 terms the entry publishes, exactly, in integer arithmetic. This is what ties the model to the entry: it is built by reading the entry's English, and a misreading gives different counts.

Second, the conjectured recurrence was evaluated directly on those published terms, with no model involved, and holds wherever the proved range and the data overlap.

Third, the derived annihilator A was itself tested on terms it had not been given: the model was evaluated beyond the S values that determine it and A reproduced them. A bound that is too small fails this test, which is why it is run.

References

[1] The OEIS Foundation, *The On-Line Encyclopedia of Integer Sequences*, https://oeis.org, 2026. Sequence A063116.

[2] M. Beck and S. Robins, *Computing the Continuous Discretely*, 2nd ed., Springer, 2015. (Ehrhart quasi-polynomials and their periods.)

[3] R. P. Stanley, *Enumerative Combinatorics*, Volume 1, 2nd ed., Cambridge University Press, 2011.

[4] F. Diamond and J. Shurman, *A First Course in Modular Forms*, Springer GTM 228, 2005.